

Matrix discrepancy and optimization

# The Komlós discrepancy conjecture

**Difficulty:** extreme **Importance:** broadly interesting**Status:** Open**Provenance:** source-stated conjecture.**Rating rationale:** Dimension-free vector balancing is a foundational discrepancy barrier with wide implications for rounding, optimization and algorithms.Does a finite constant  $C > 0$  exist such that, for every pair of positive integers  $m, n$  and every matrix  $A = (a_{ij}) \in \mathbb{R}^{m \times n}$  satisfying

$$\sum_{i=1}^m a_{ij}^2 \leq 1 \quad (1 \leq j \leq n),$$

there is a vector  $x \in \{-1, 1\}^n$  for which

$$\|Ax\|_\infty = \max_{1 \leq i \leq m} \left| \sum_{j=1}^n a_{ij} x_j \right| \leq C?$$

The same constant must work for every dimension and matrix. The question asks for existence of a signing; it does not additionally require an algorithm.

This is a matrix balancing problem: choose column signs so that their sum is small in every coordinate. Its norm constraint and simultaneous coordinate control connect discrepancy theory with rounding and linear optimization.

## References

1. N. Bansal and H. Jiang, *An Exposition of the  $\tilde{O}(\log^{1/4} n)$  Bound for the Komlós Problem*, arXiv:2608.28452 (2026), abstract and introduction, including the explicit conjecture and the new dimension-dependent bound. [Paper](#).
2. N. Bansal and H. Jiang, *Decoupling via Affine Spectral-Independence: Beck-Fiala and Komlós Bounds Beyond Banaszczyk*, arXiv:2508.03961v2 (2025), abstract and introductory results. [Paper](#).
3. W. Banaszczyk, *Balancing vectors and Gaussian measures of  $n$ -dimensional convex bodies*, Random Structures & Algorithms 12(4) (1998), pp. 351–360, the main vector-balancing theorem underlying the earlier logarithmic bound. [Article](#).

## Status check — 2026-09-10

checked the current arXiv records (2608.28452v1, 2026-08-28; 2508.03961v2, 2025-09-09) and searched “Komlos conjecture solved proof September 2026” and “Komlos Bansal Jiang 2026 bound”. The new bound is  $O((\log n)^{1/4}(\log \log n)^{7/4})$  in its asymptotic range, not a universal constant. No full proof, counterexample, or withdrawal of these source papers was found. The disproved Hajela conjecture and the separate matrix Spencer conjecture are not this statement.**Audit update (2026-09-10):** Rechecked the August 2026 exposition and searched for later Komlós resolutions. Its improved dimension-dependent bound still leaves the universal-constant target open. This is a bounded literature check, not a proof that no solution exists.